

FACULTÉ DES SCIENCES ET TECHNOLOGIES

(FST)

TROISIÈME ANNÉE

Rapport du travail de Laboratoire Nº1

Cours : Systèmes

Étudiant : Mikëlo GONTRAND

Professeur : Ismaël SAINT AMOUR

LE 23 OCTOBRE 2025.

L’objectif de ce TD est de :

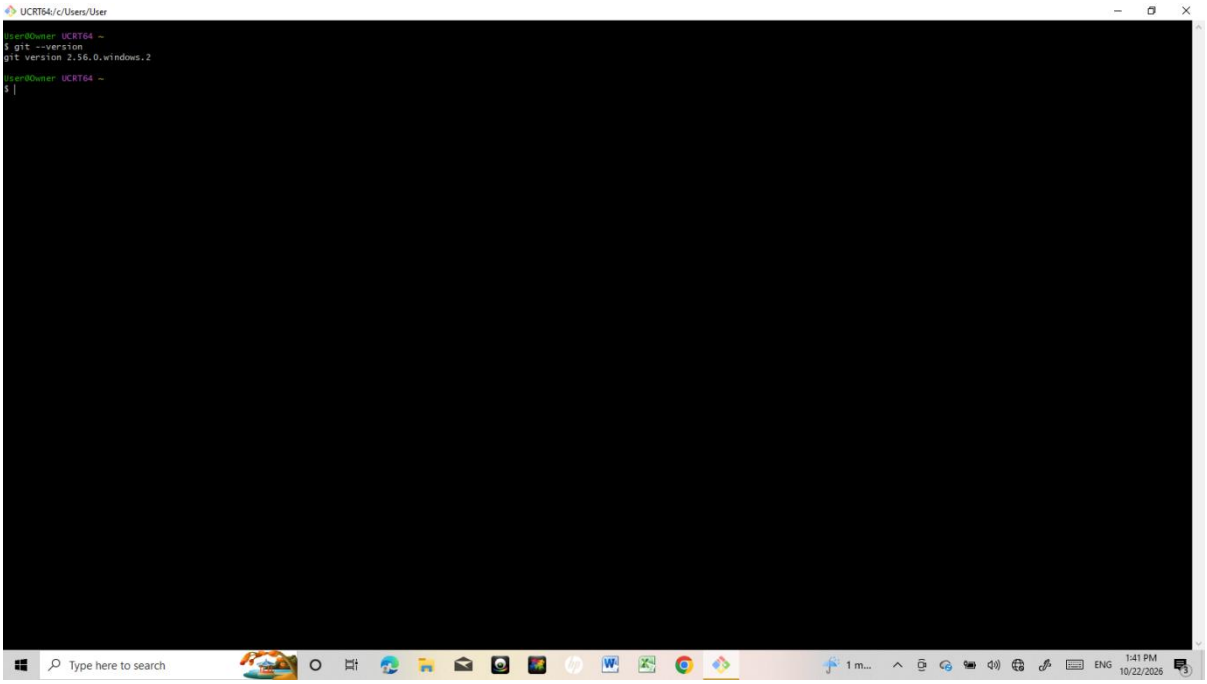
1) Comprendre les concepts de base de Git (système de gestion de versions).
2) Savoir utiliser Git pour gérer un projet localement.
3) Découvrir Github et apprendre à collaborer sur un dépôt distant.

Travail Dirigé

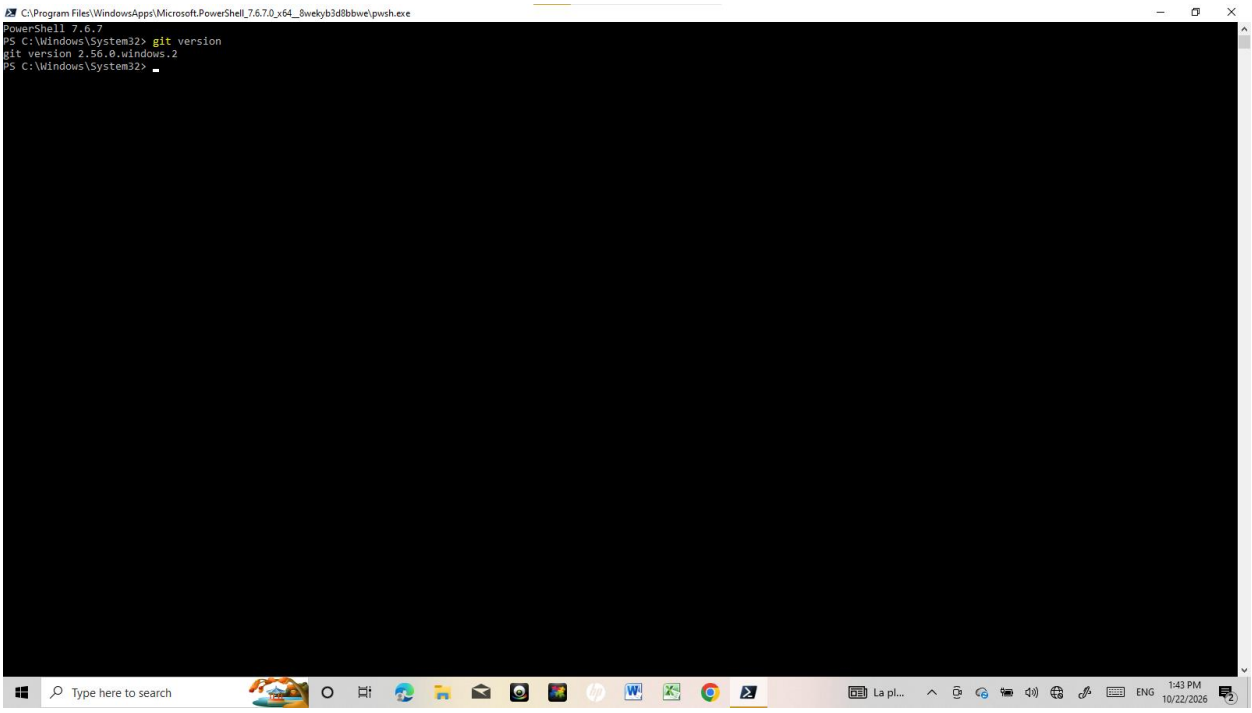
1) Reproduire une série de tâches.
2) Crée un nouveau dépôt sur Github.

1) Vérification de la Git et PowerShell installée sur le PC.

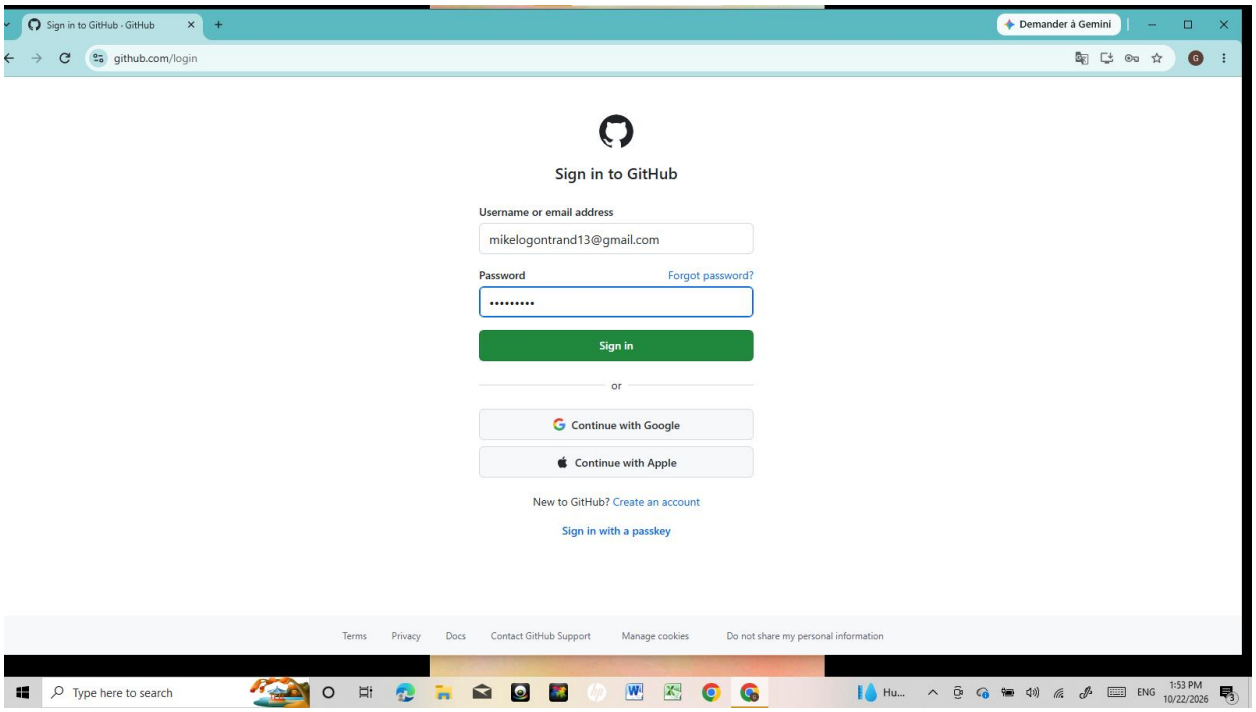
Git --
version

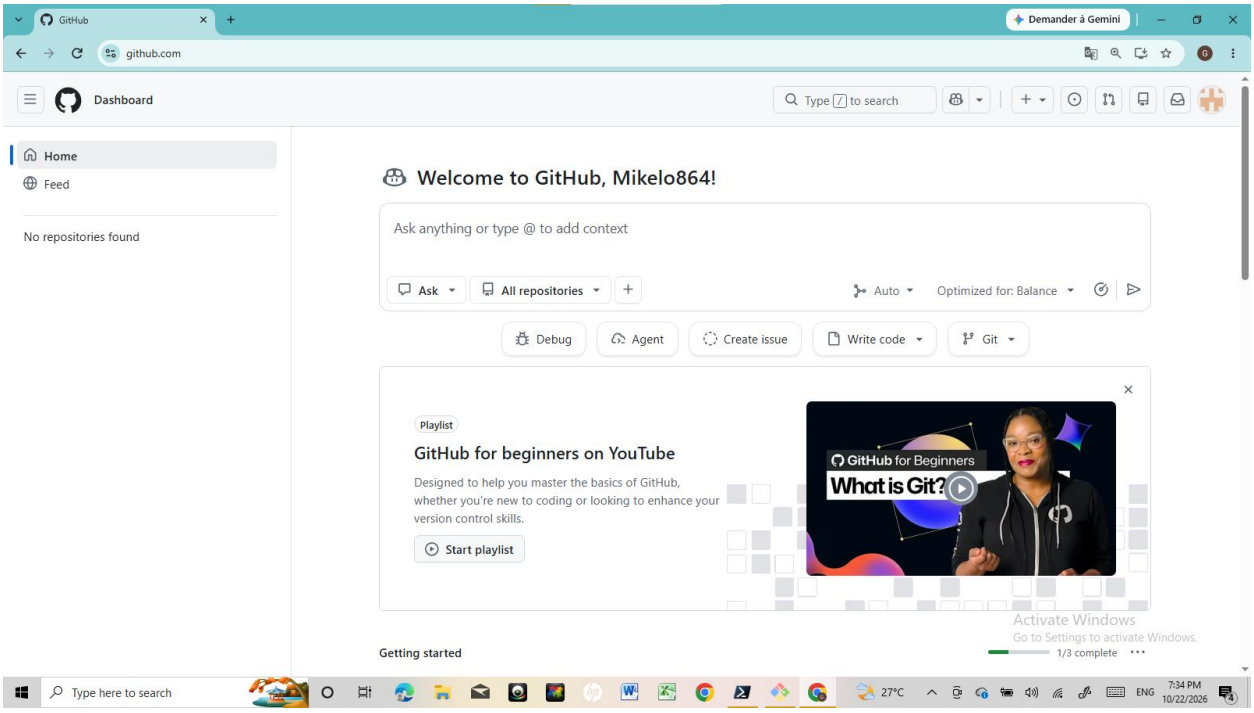


Git version



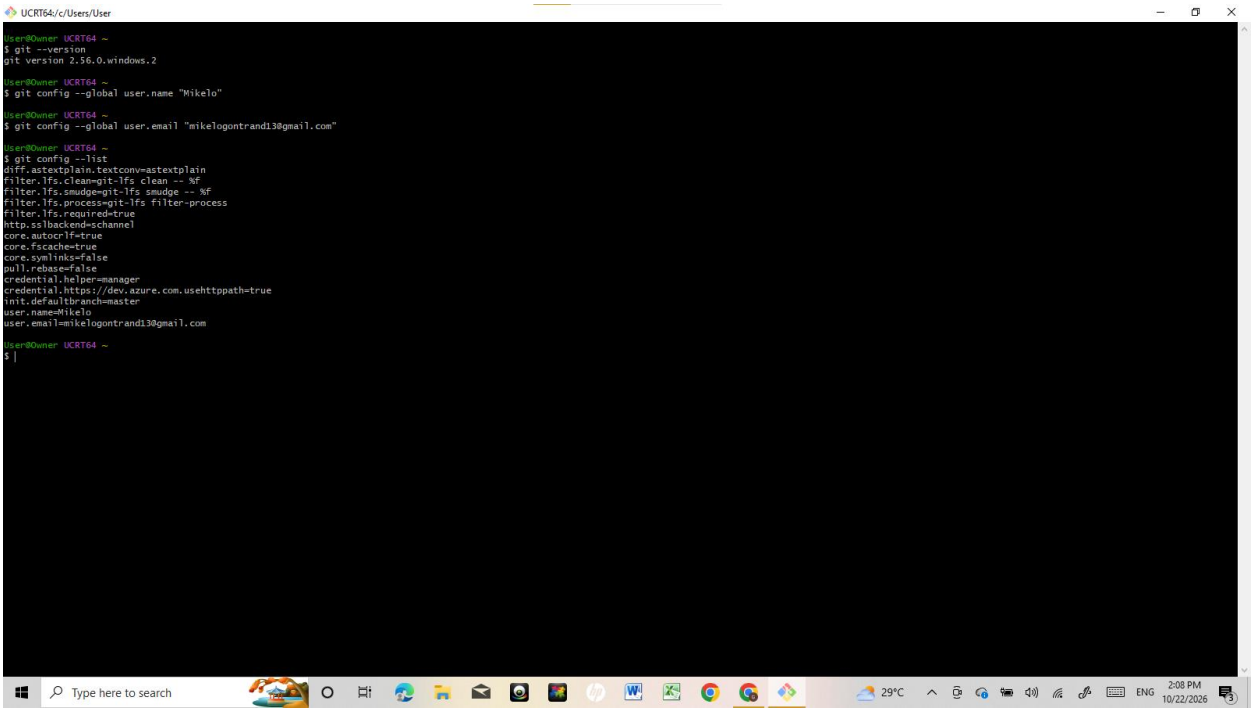
2) Connection à mon compt Github.





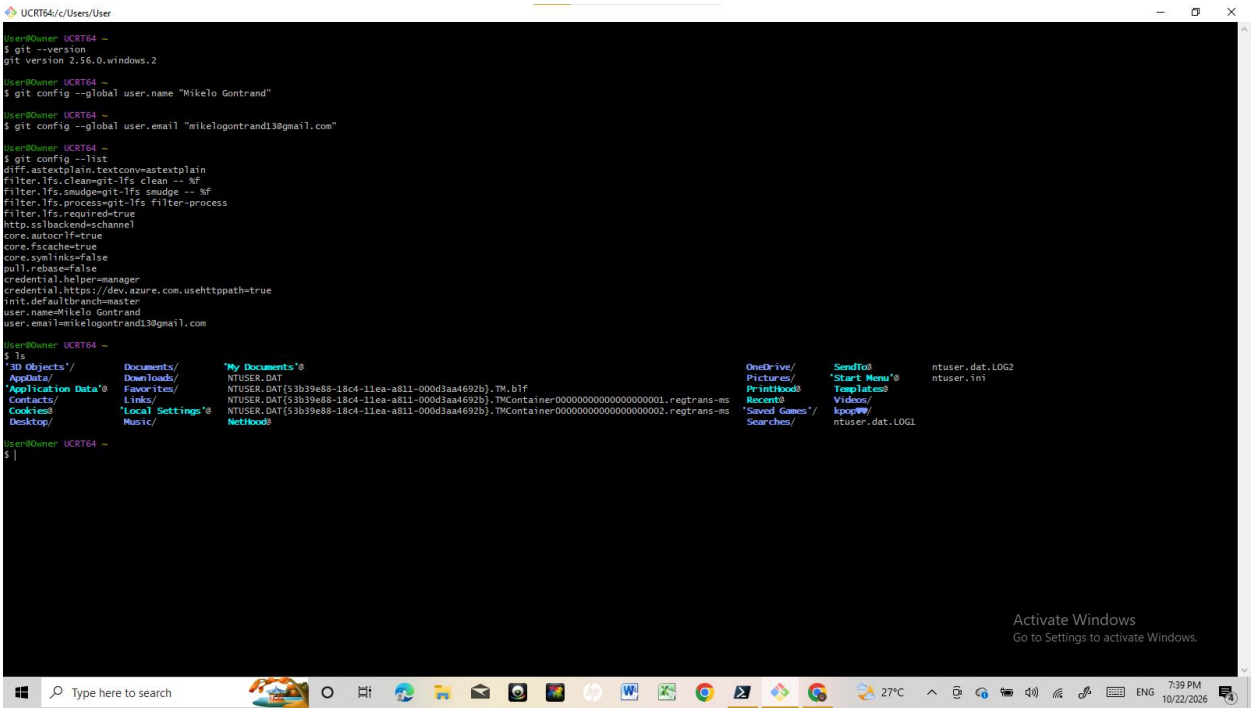
3) Configuration de Git et vérification des données utilisateurs.

git config --global user.name
git config --global user.email
git config --list



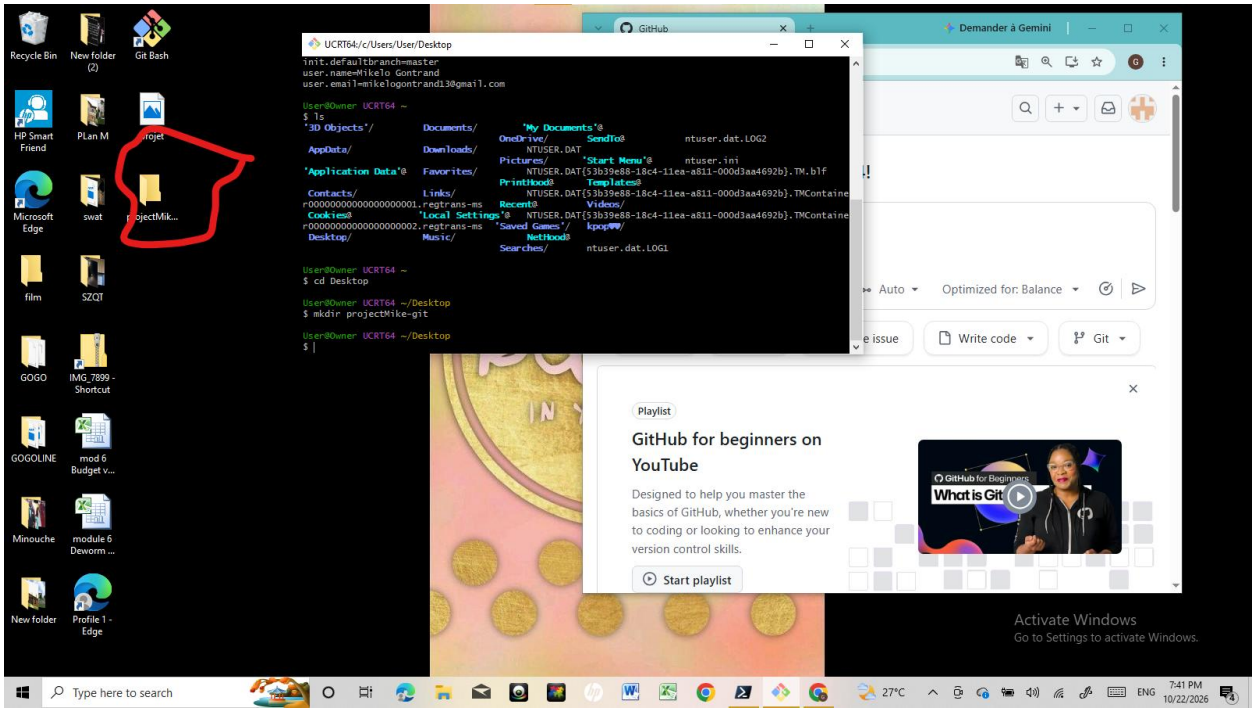
4) Affichage de la liste des fichiers et dossiers avec la commande :

Ls



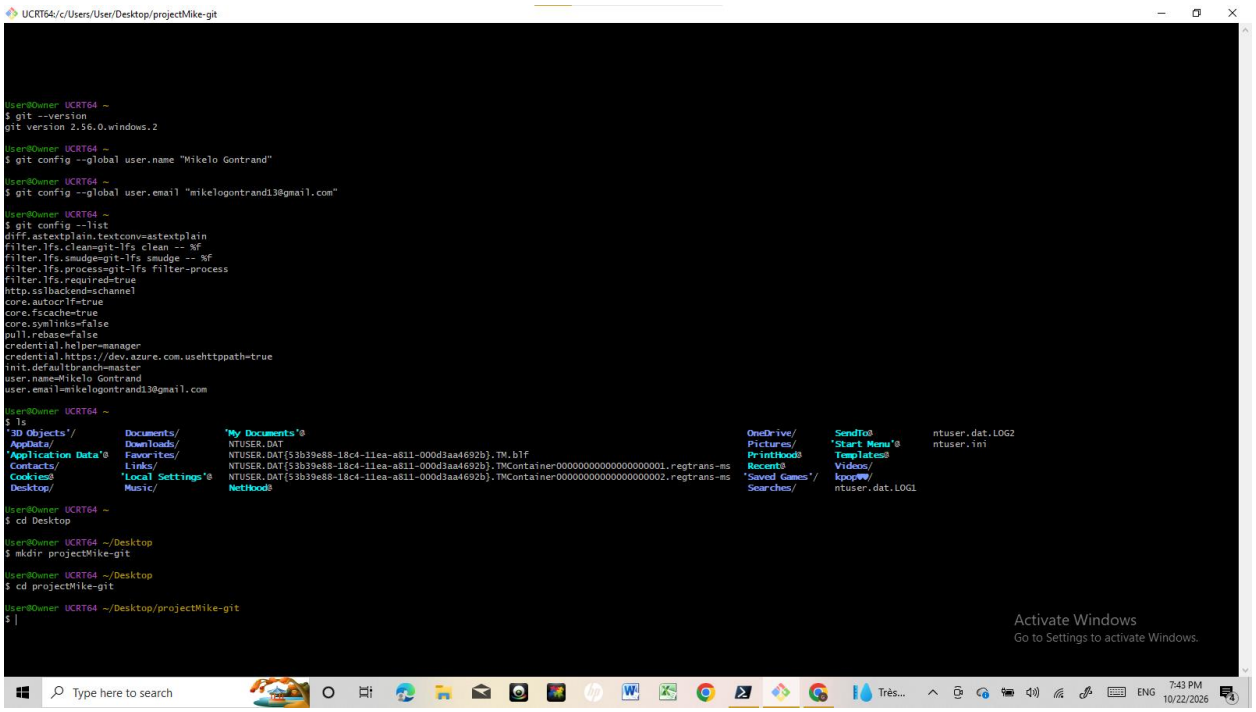
5) Création d'un dossier dans Desktop

Cd Desktop
Mkdir projectMike-git



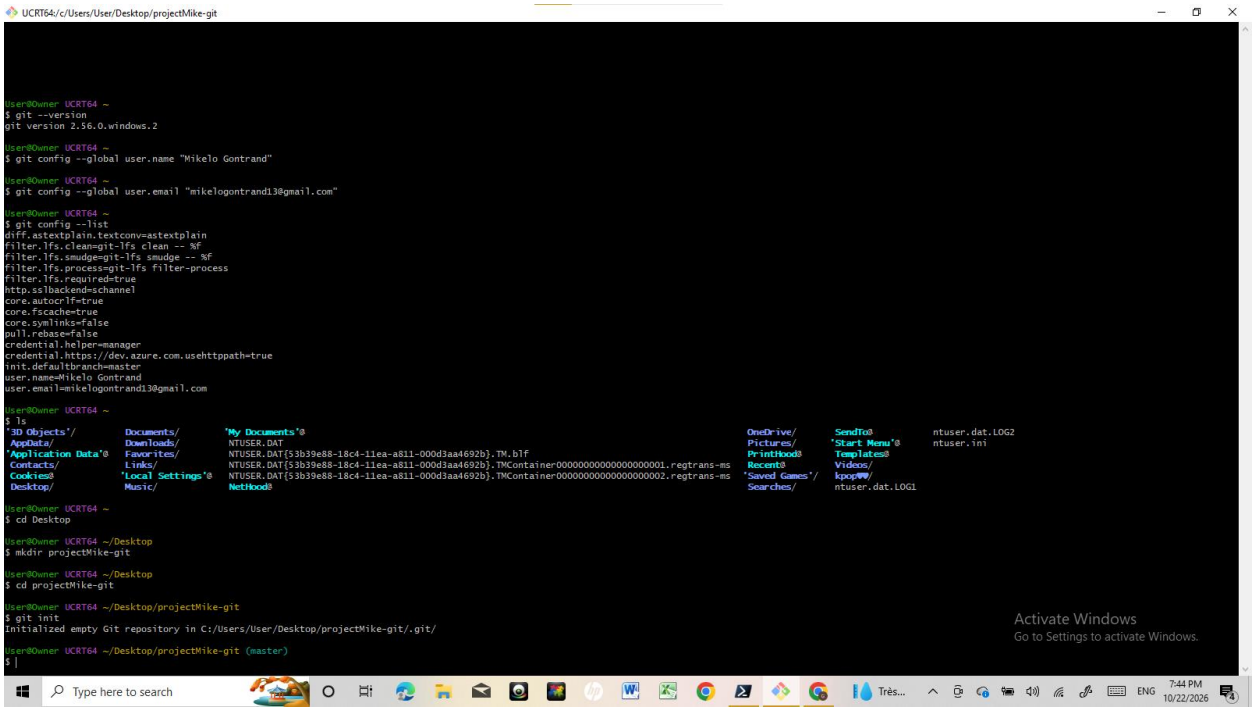
6) Entrée dans notre dossier << projectMike-git>> qui vient de créer.

Cd projectMike-git



7) Initialisation du dépôt git

git init

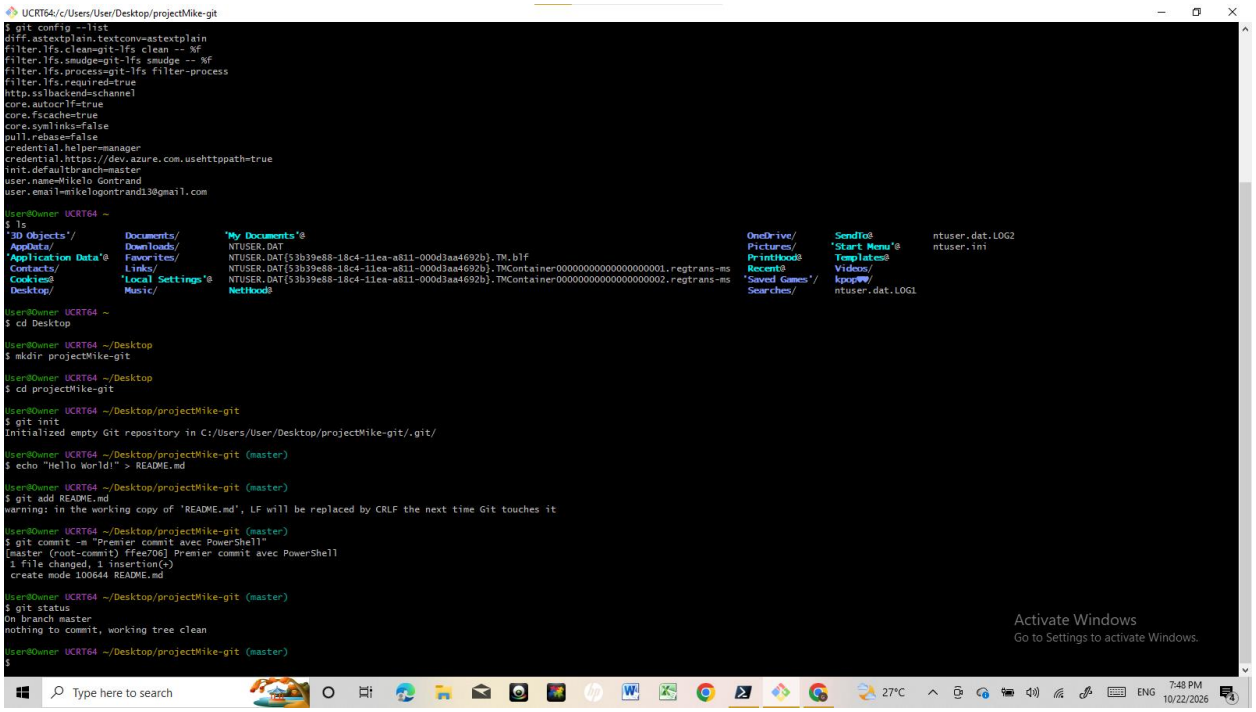


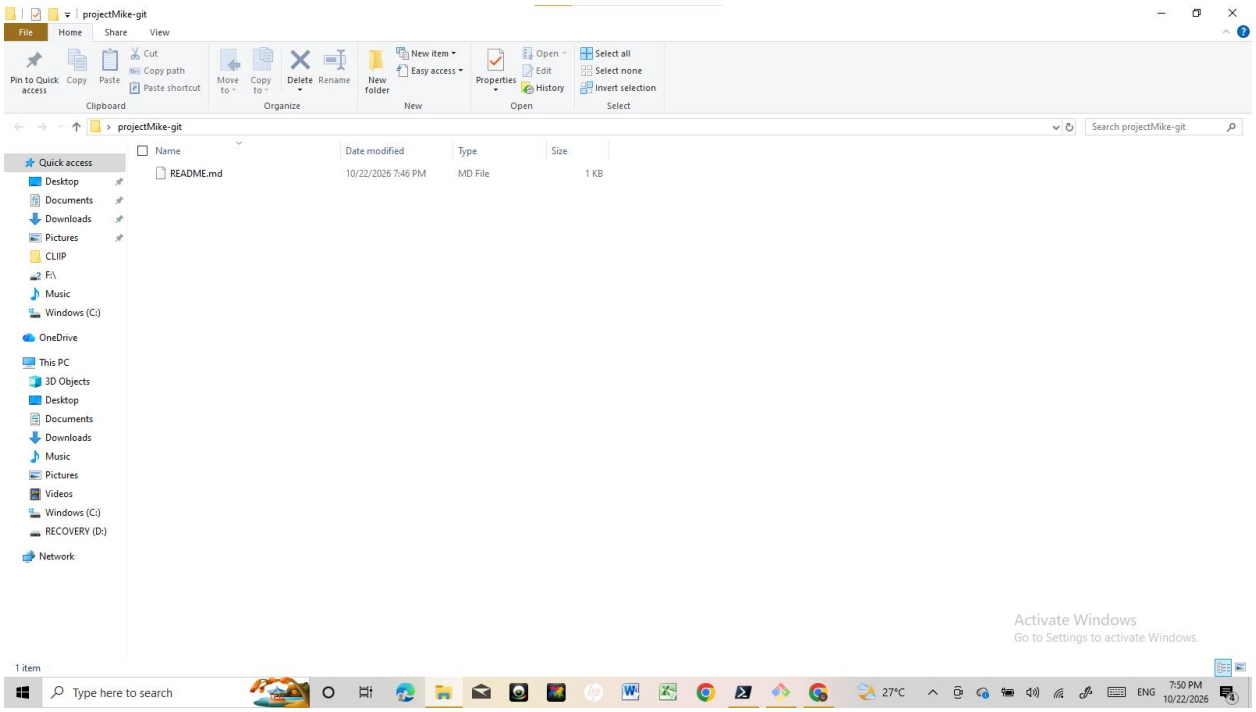
8) Ouverture et écriture de mon projet dans le fichier README.md ainsi ajout du dossier README.md et premier commit effectué.

echo “ Hello World” . README.md

git commit –m “ Premier commit avec PowerShell”

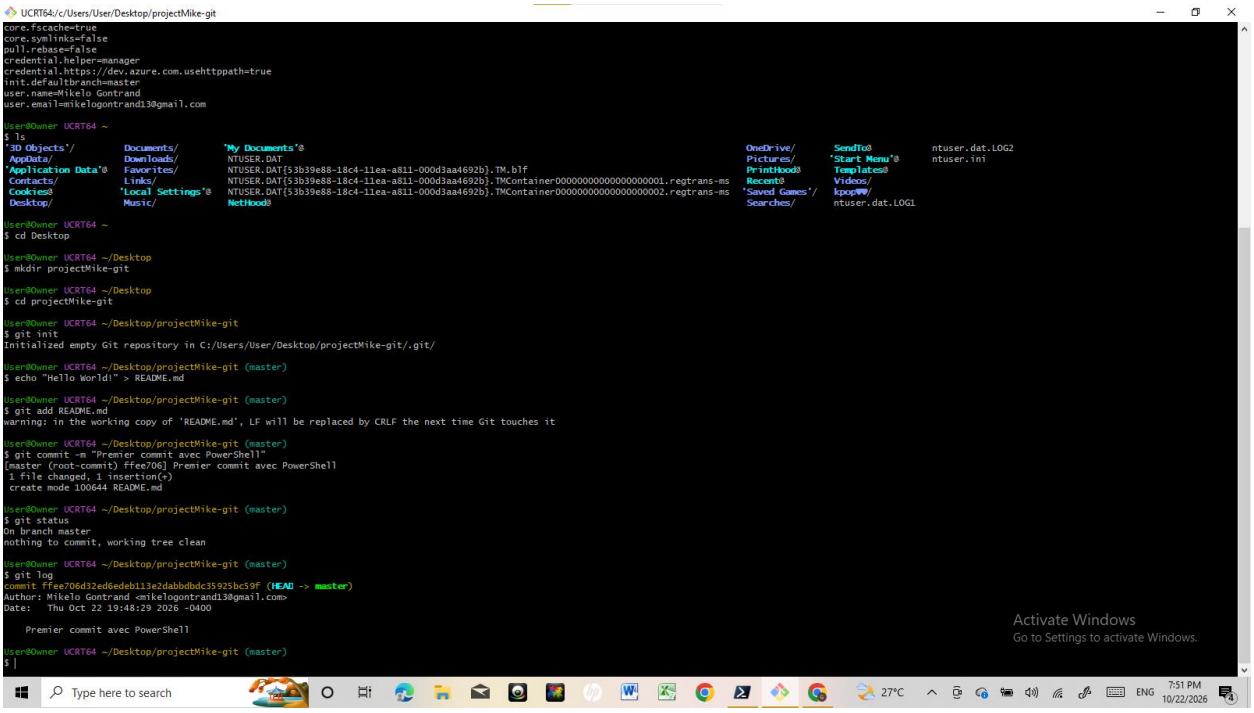
git status





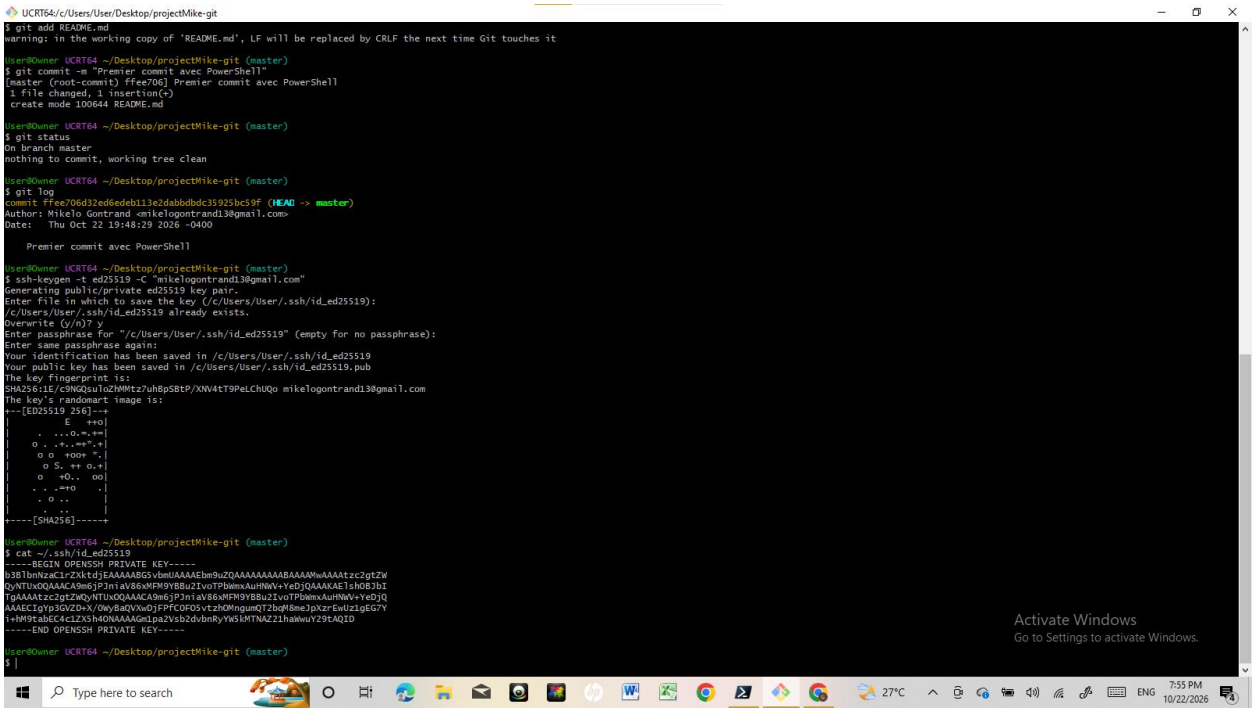
9) Affichage de l'historique des commits dans un dépôt Git.

git log



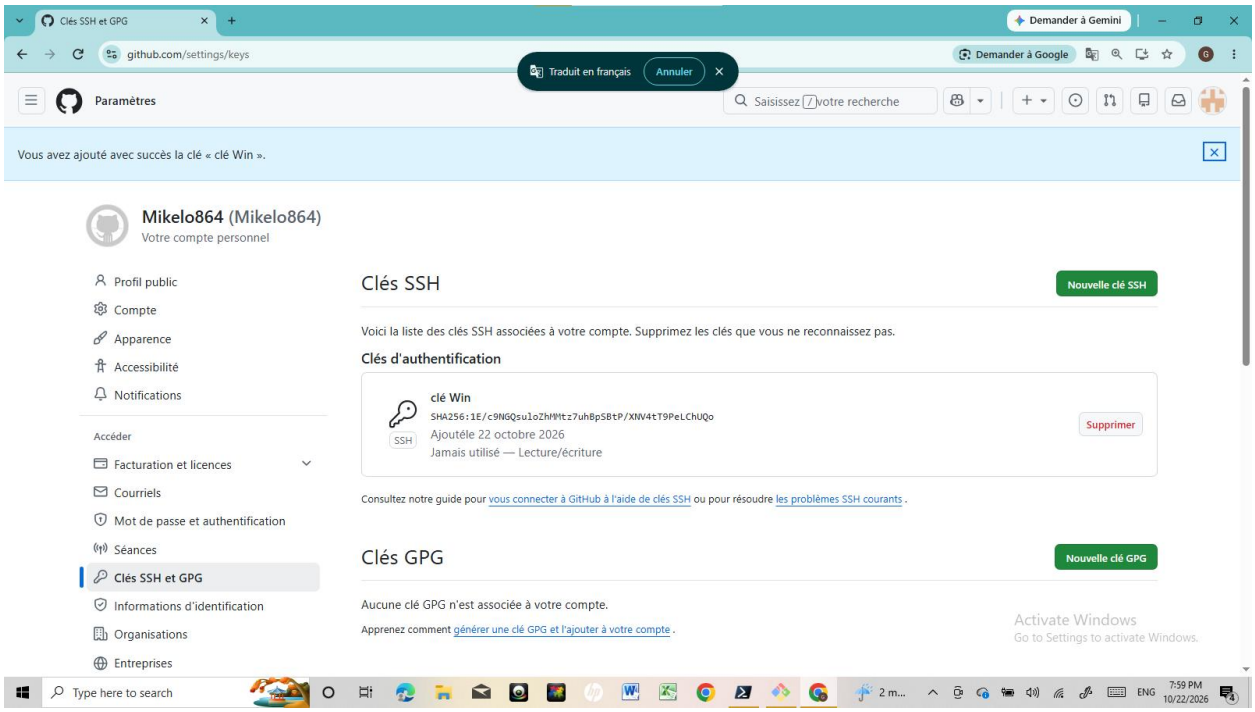
10) Génération d'une clé SSH et affichage de la clé privée.

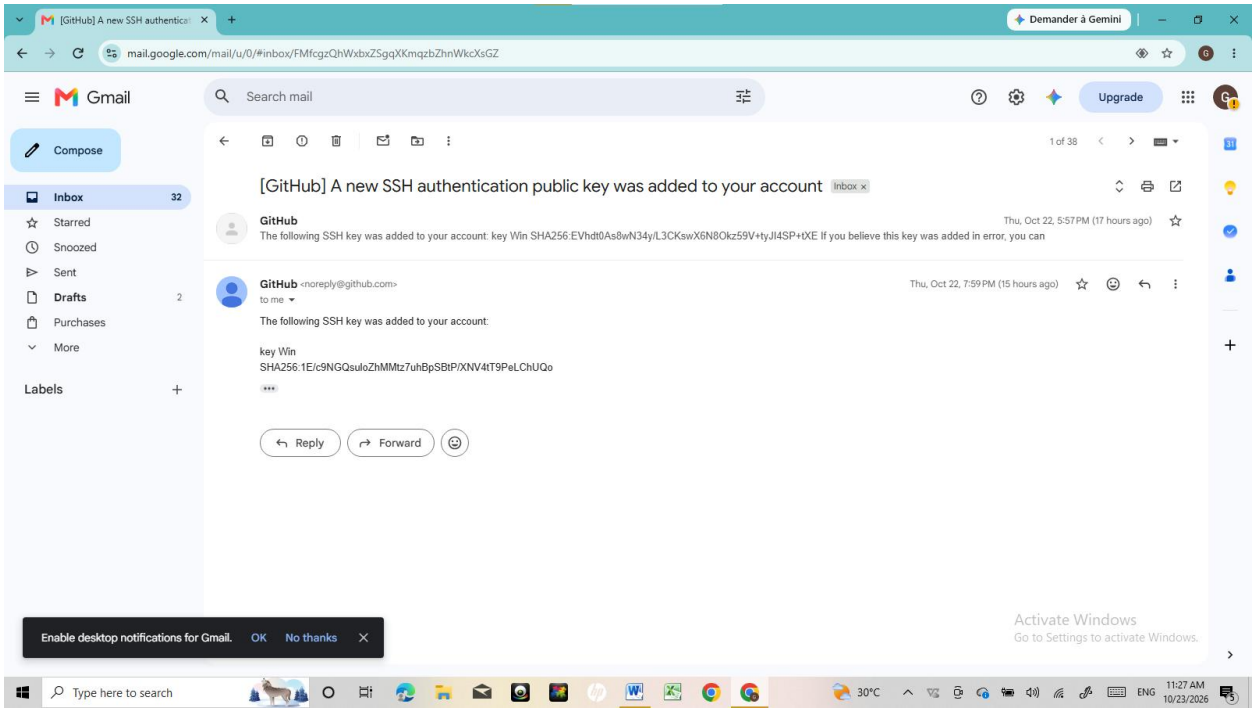
ssh-keygen –t ed25519 –C « mikelogontrand13@gmail.com »
cat ~/.ssh/id_ed25519



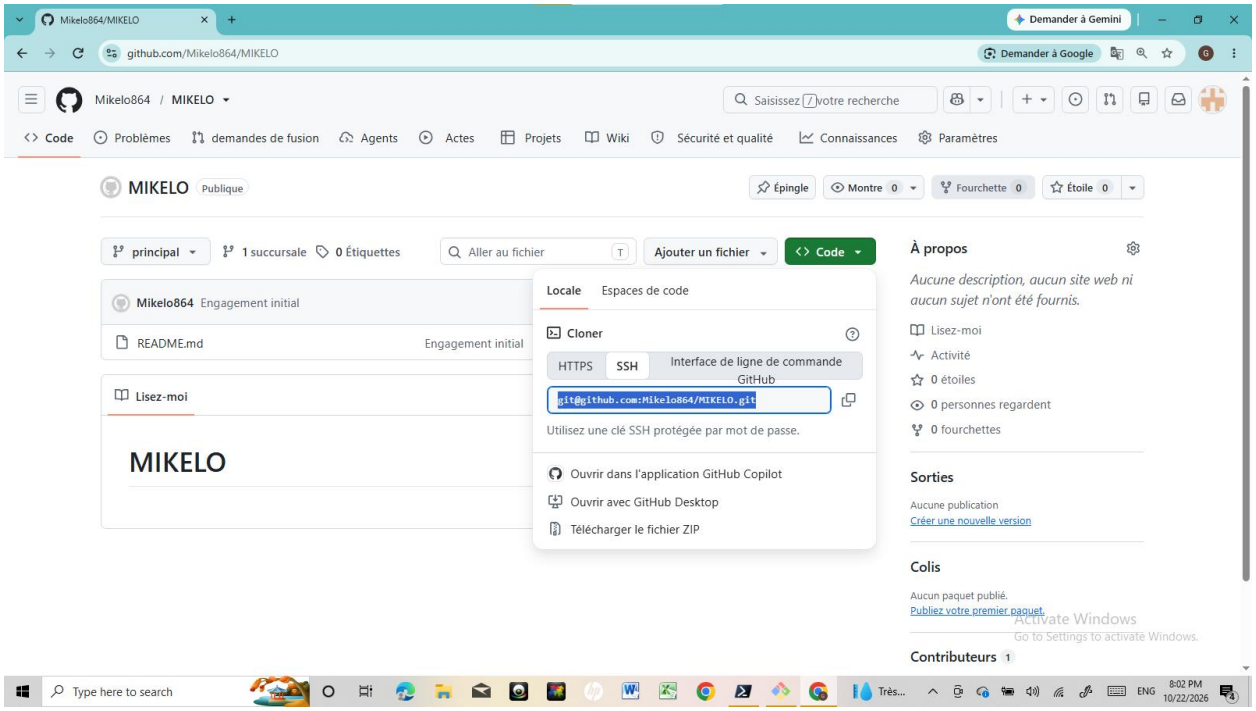
11) Affichage de la clé publique.

cat ~/.ssh/id_ed25519.pub

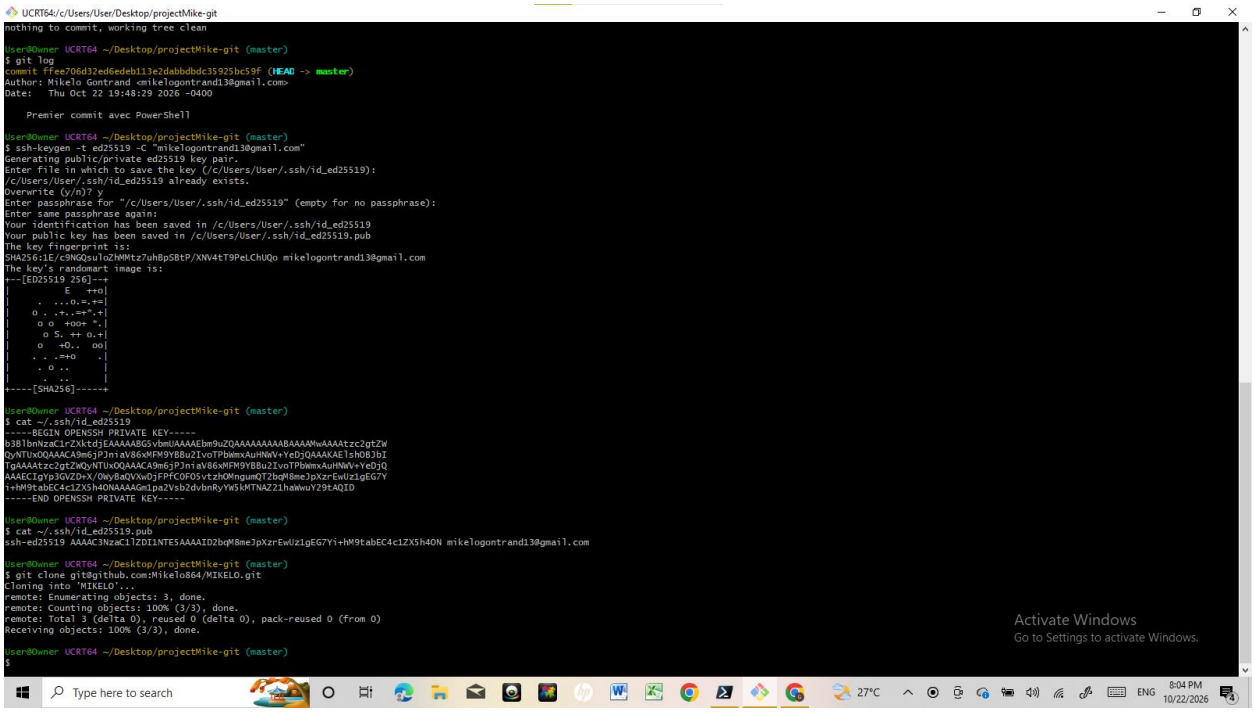




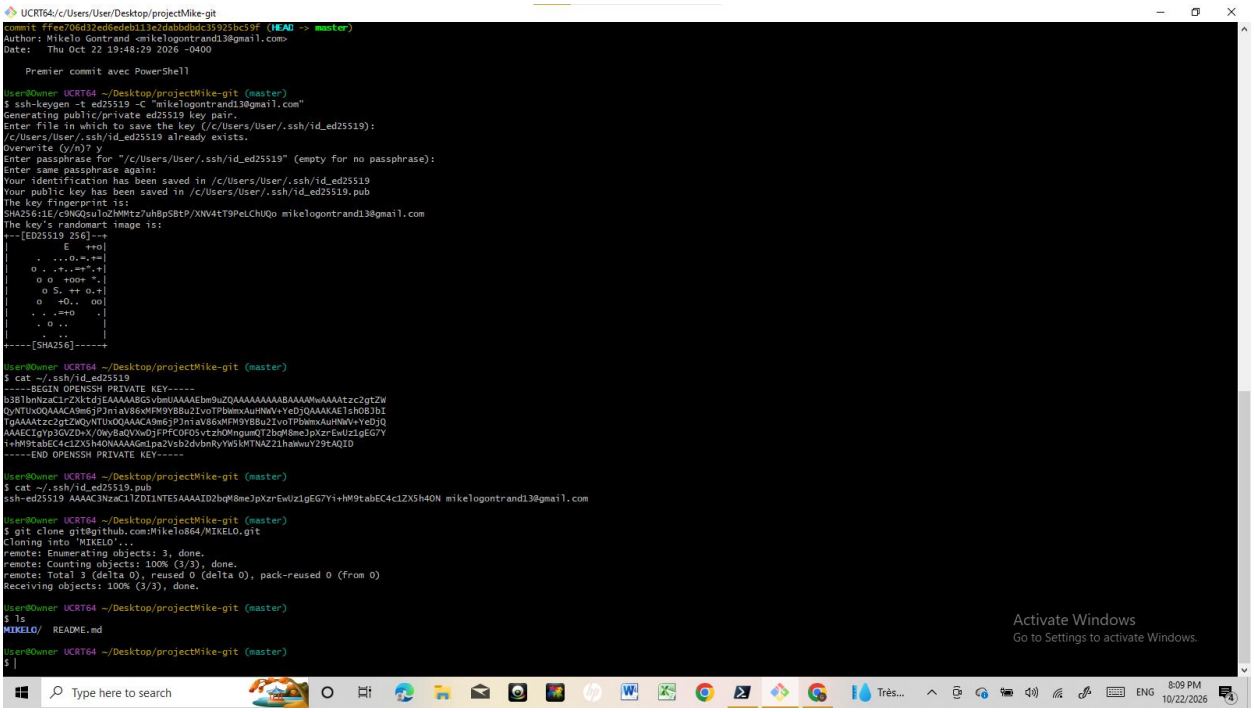
12) Clonage du dépôt Github



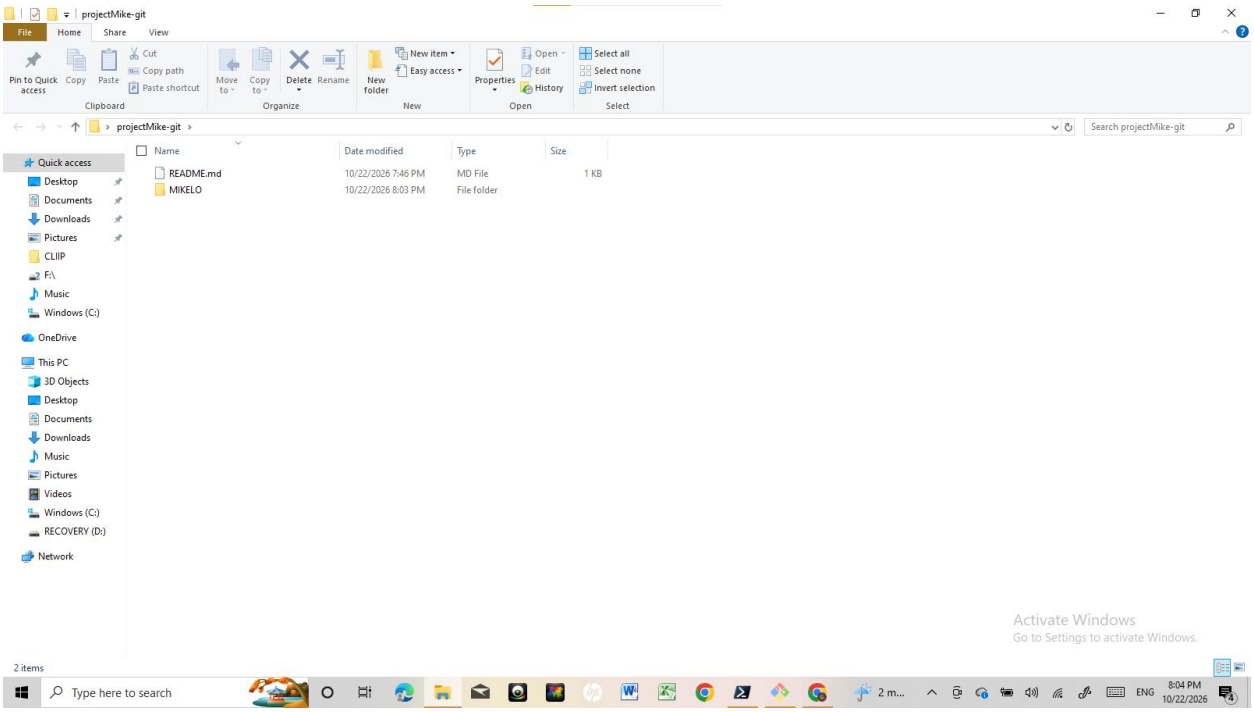
git clone git@github.com:Mikelo864/MIKELO.git


UCRT64:/c/Users/User/Desktop/projectMike-git
nothing to commit, working tree clean
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ git log
commit ffee706d32ed6edeb113e2dabbdbdc35925bc59f (HEAD -> master)
Author: Mikelo Gontrand <mikelogontrand13@gmail.com>
Date: Thu Oct 22 19:48:29 2026 -0400
Premier commit avec PowerShell
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ ssh-keygen -t ed25519 -C "mikelogontrand13@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/User/.ssh/id_ed25519):
/c/Users/User/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase for "/c/Users/User/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/User/.ssh/id_ed25519
Your public key has been saved in /c/Users/User/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:1E/c9NGQsuloZhMMtz7uhBpSBtP/XNV4tT9PeLChUQo mikelogontrand13@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
+----[SHA256]-----+
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ cat ~/.ssh/id_ed25519
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAAAMwAAAAtzc2gtZW
QyNTUxOQAAACA9m6jPJniaV86xMFM9YBBu2IvoTPbWmxAuHNWV+YeDjQAAAKAIshOBJbI
TgAAAAtzc2gtZWQyNTUxOQAAACA9m6jPJniaV86xMFM9YBBu2IvoTPbWmxAuHNWV+YeDjQ
AAAECIgYp3GVZD+X/OWyBaQVXwDjFPfCOFO5vtzhOMngumQT2bqM8meJpXzrEwUz1gEG7Y
i+hM9tabEC4c1ZX5h4ONAAAAGm1pa2Vsb2dvbnRyYW5kMTNAZ21haWwuY29tAQID
-----END OPENSSH PRIVATE KEY-----
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAID2bqM8meJpXzrEwUz1gEG7Yi+hM9tabEC4c1ZX5h4ON mikelogontrand13@gmail.com
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ git clone git@github.com:Mikelo864/MIKELO.git
Cloning into 'MIKELO'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$
Activate Windows
Go to Settings to activate Windows.
Type here to search
27°C
ENG
8:04 PM
10/22/2026

ls


UCRT64:/c/Users/User/Desktop/projectMike-git
commit ffee706d32ed6edeb113e2dabbdbdc35925bc59f (HEAD -> master)
Author: Mikelo Gontrand <mikelogontrand13@gmail.com>
Date: Thu Oct 22 19:48:29 2026 -0400
Premier commit avec PowerShell
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ ssh-keygen -t ed25519 -C "mikelogontrand13@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/User/.ssh/id_ed25519):
/c/Users/User/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase for "/c/Users/User/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/User/.ssh/id_ed25519
Your public key has been saved in /c/Users/User/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:1E/c9NGQsuloZhMMtz7uhBpSBtP/XNV4tT9PeLChUQo mikelogontrand13@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
+----[SHA256]-----+
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ cat ~/.ssh/id_ed25519
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAAAMwAAAAtzc2gtZW
QyNTUxOQAAACA9m6jPJniaV86xMFM9YBBu2IvoTPbWmxAuHNWV+YeDjQAAAKAIshOBJbI
TgAAAAtzc2gtZWQyNTUxOQAAACA9m6jPJniaV86xMFM9YBBu2IvoTPbWmxAuHNWV+YeDjQ
AAAECIgYp3GVZD+X/OWyBaQVXwDjFPfCOFO5vtzhOMngumQT2bqM8meJpXzrEwUz1gEG7Y
i+hM9tabEC4c1ZX5h4ONAAAAGm1pa2Vsb2dvbnRyYW5kMTNAZ21haWwuY29tAQID
-----END OPENSSH PRIVATE KEY-----
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAID2bqM8meJpXzrEwUz1gEG7Yi+hM9tabEC4c1ZX5h4ON mikelogontrand13@gmail.com
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ git clone git@github.com:Mikelo864/MIKELO.git
Cloning into 'MIKELO'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$ ls
MIKELO/ README.md
User@Owner UCRT64 ~/Desktop/projectMike-git (master)
$
Activate Windows
Go to Settings to activate Windows.
Type here to search
Très...
ENG
8:09 PM
10/22/2026

Dans le projet MIKELO j’ajout pas licence.



cd projectMikelo

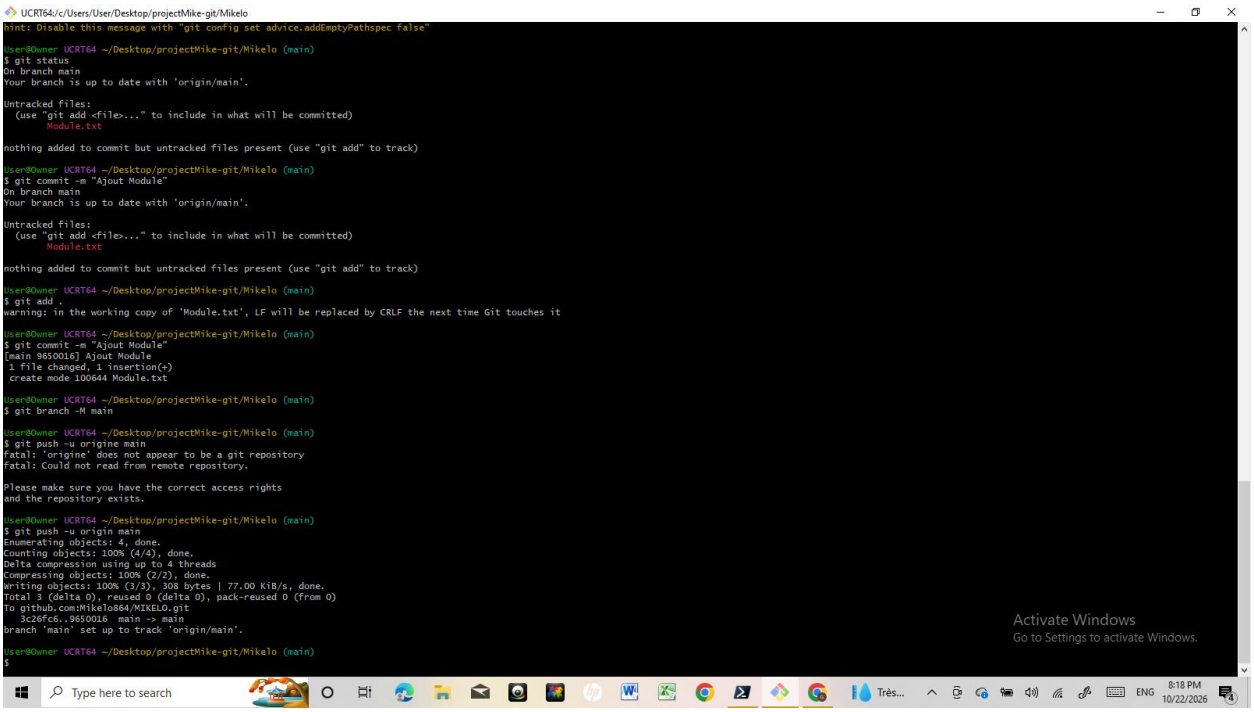
echo « système d’exploitation ! » > Module.txt

git add

git commit –m « Ajout Module »

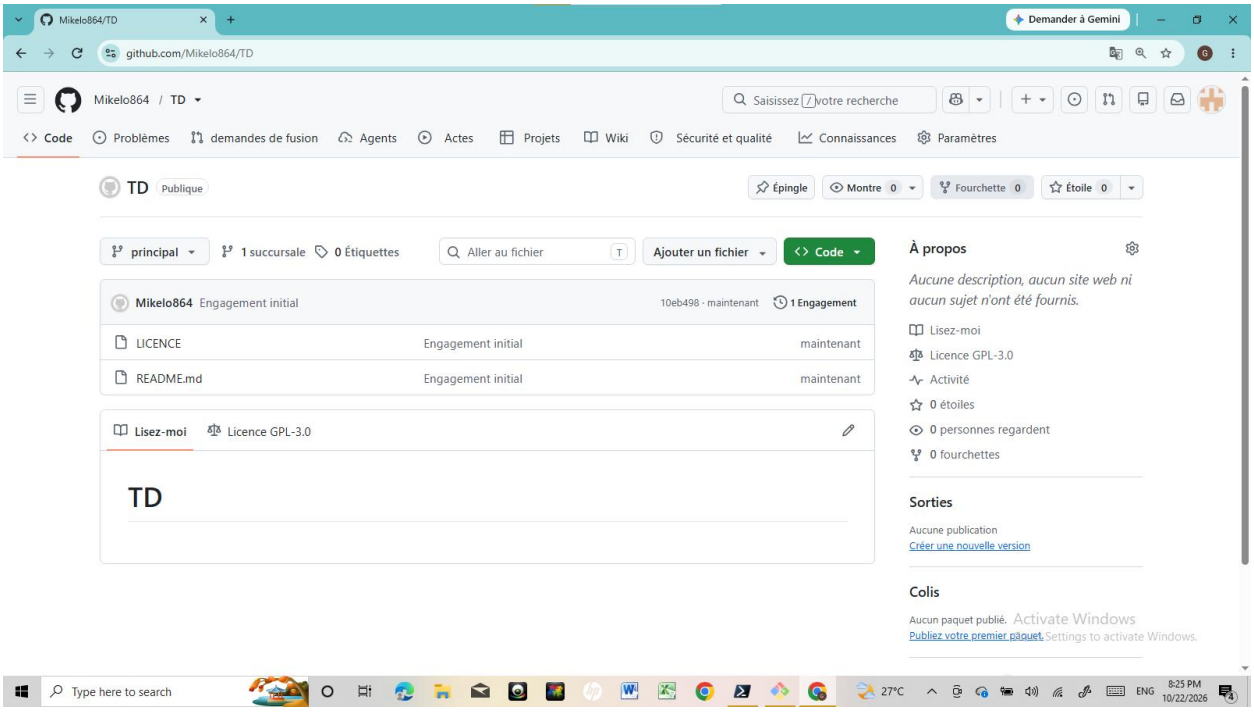
git branch –M main

git push –u origin main


UCRT64:/c/Users/User/Desktop/projectMike-git/Mikelo
hint: Disable this message with "git config set advice.addEmptyPathspec false"
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.
Untracked files:
(use "git add <file>..." to include in what will be committed)
Module.txt
nothing added to commit but untracked files present (use "git add" to track)
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ git commit -m "Ajout Module"
On branch main
Your branch is up to date with 'origin/main'.
Untracked files:
(use "git add <file>..." to include in what will be committed)
Module.txt
nothing added to commit but untracked files present (use "git add" to track)
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ git add .
warning: in the working copy of 'Module.txt', LF will be replaced by CRLF the next time Git touches it
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ git commit -m "Ajout Module"
[main 9650016] Ajout Module
1 file changed, 1 insertion(+)
create mode 100644 Module.txt
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ git branch -M main
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ git push -u origine main
fatal: 'origine' does not appear to be a git repository
fatal: Could not read from remote repository.
Please make sure you have the correct access rights
and the repository exists.
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 308 bytes | 77.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Mikelo864/MIKELO.git
3c26fc6..9650016 main -> main
branch 'main' set up to track 'origin/main'.
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$
Activate Windows
Go to Settings to activate Windows.
Type here to search
Très...
ENG
8:18 PM
10/22/2026

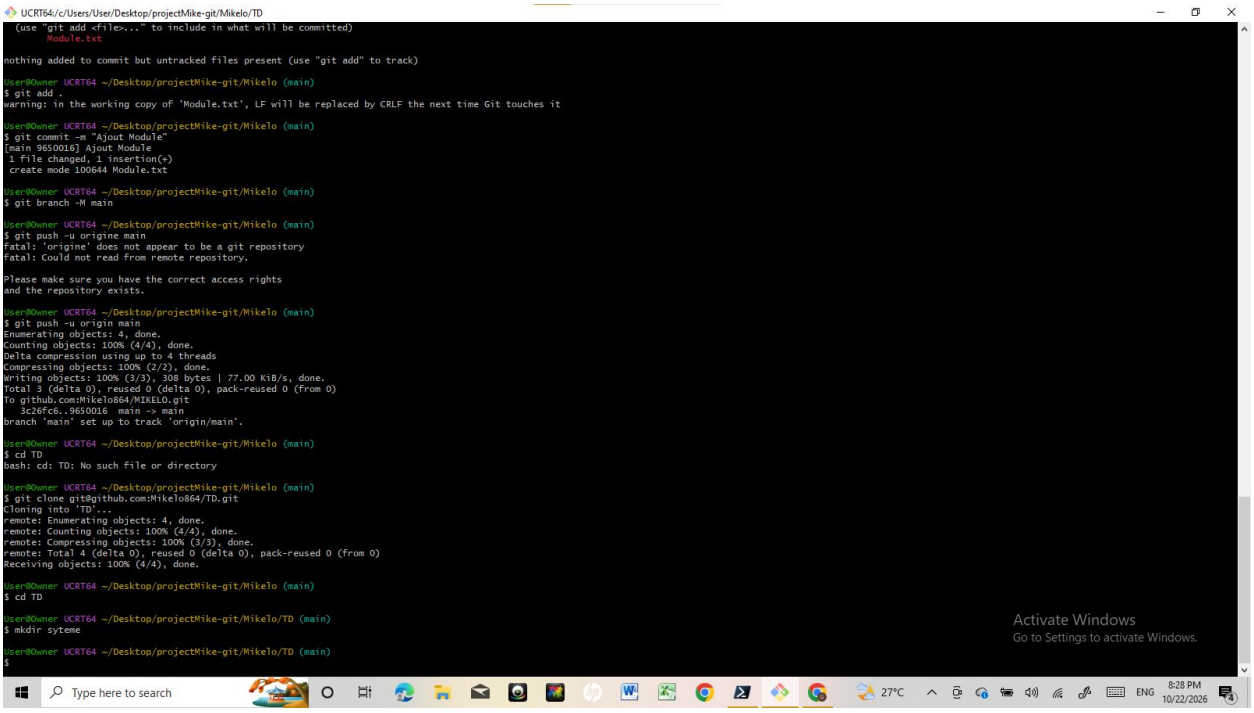
SECONDE EXERCICE

Création d’un dépôt Github nommé TD

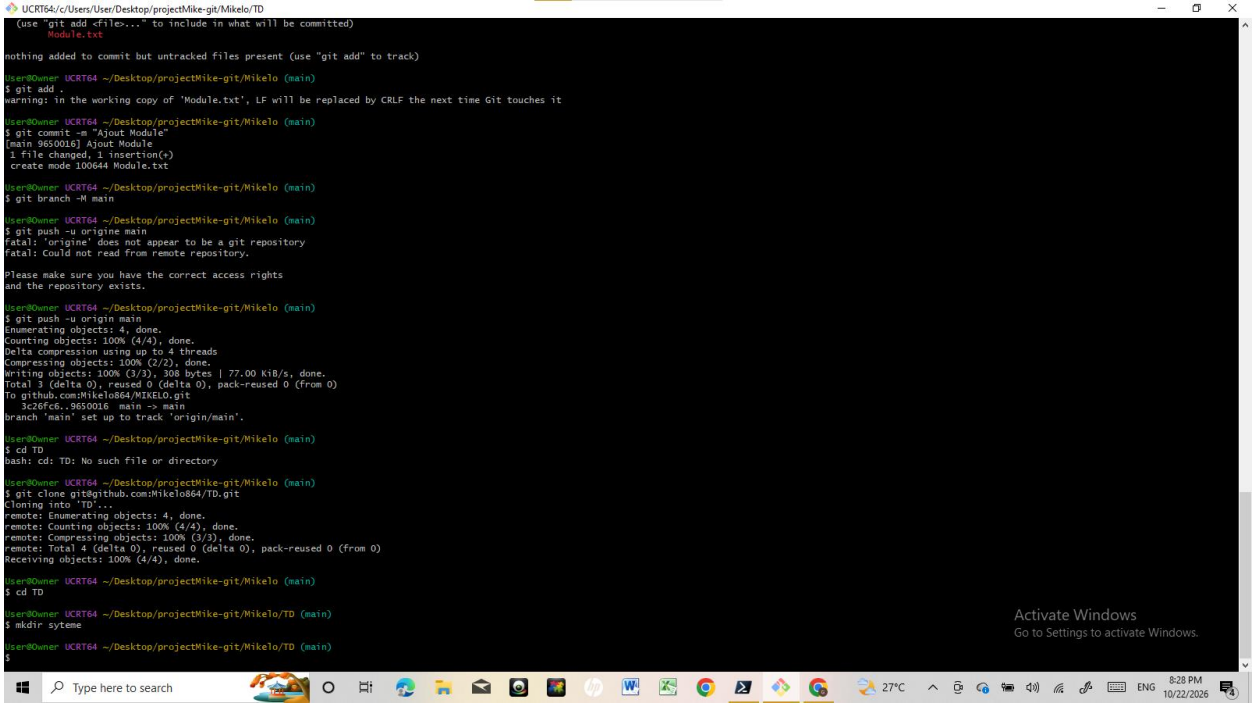


- **Clonage du dossier sur mon bureau**

git clone git@github.com:Mikelo864/TD.git



- **Création des dossiers système, image ,projet, presentation dans le dossier principal TD.**



cd TD

mkdir système

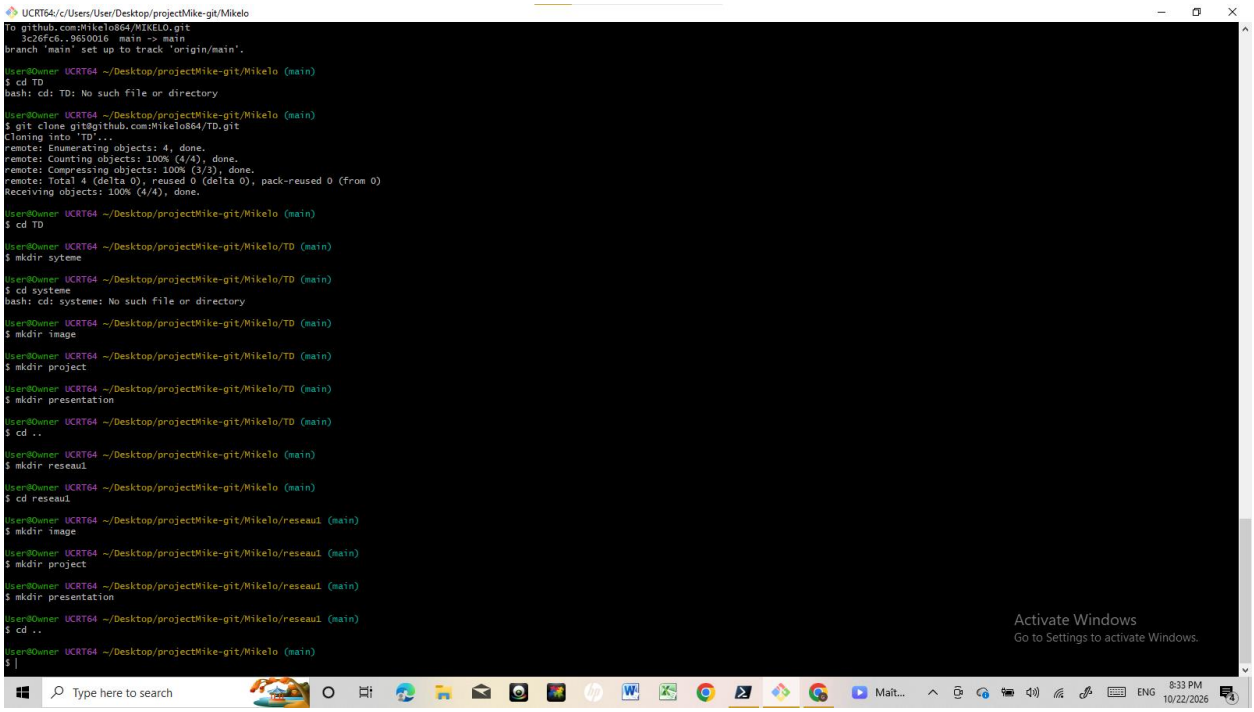
cd système

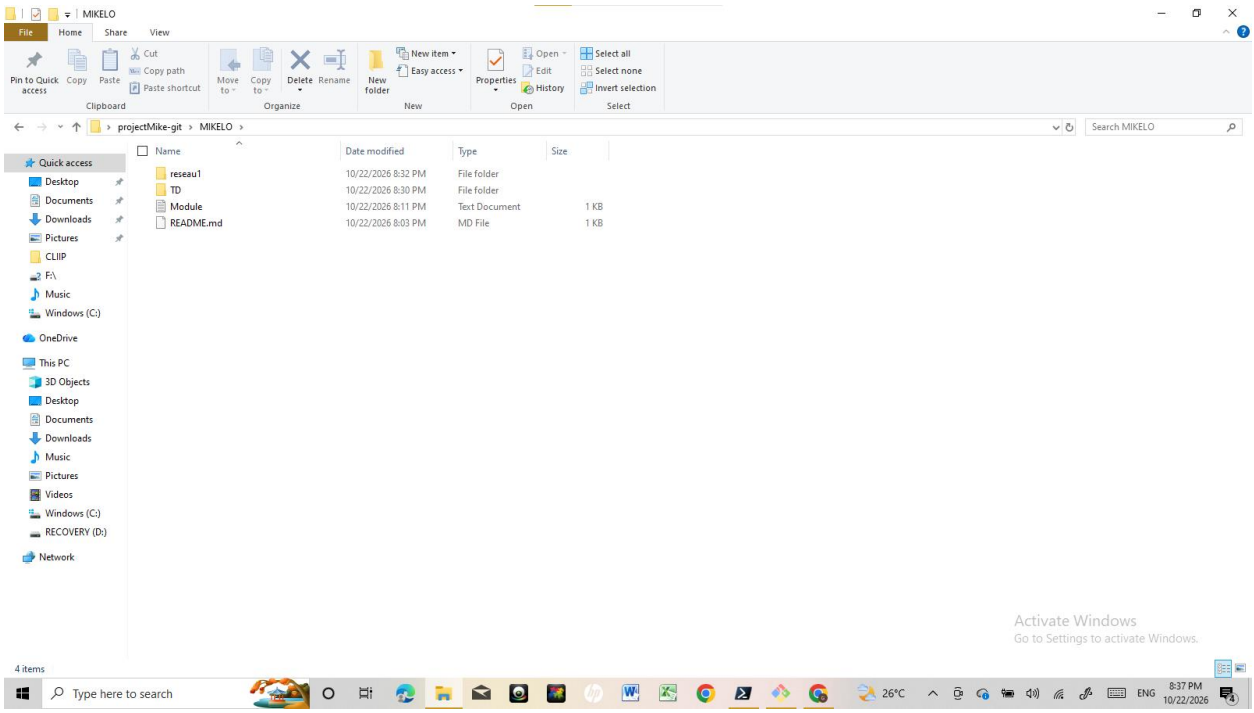
mkdir image

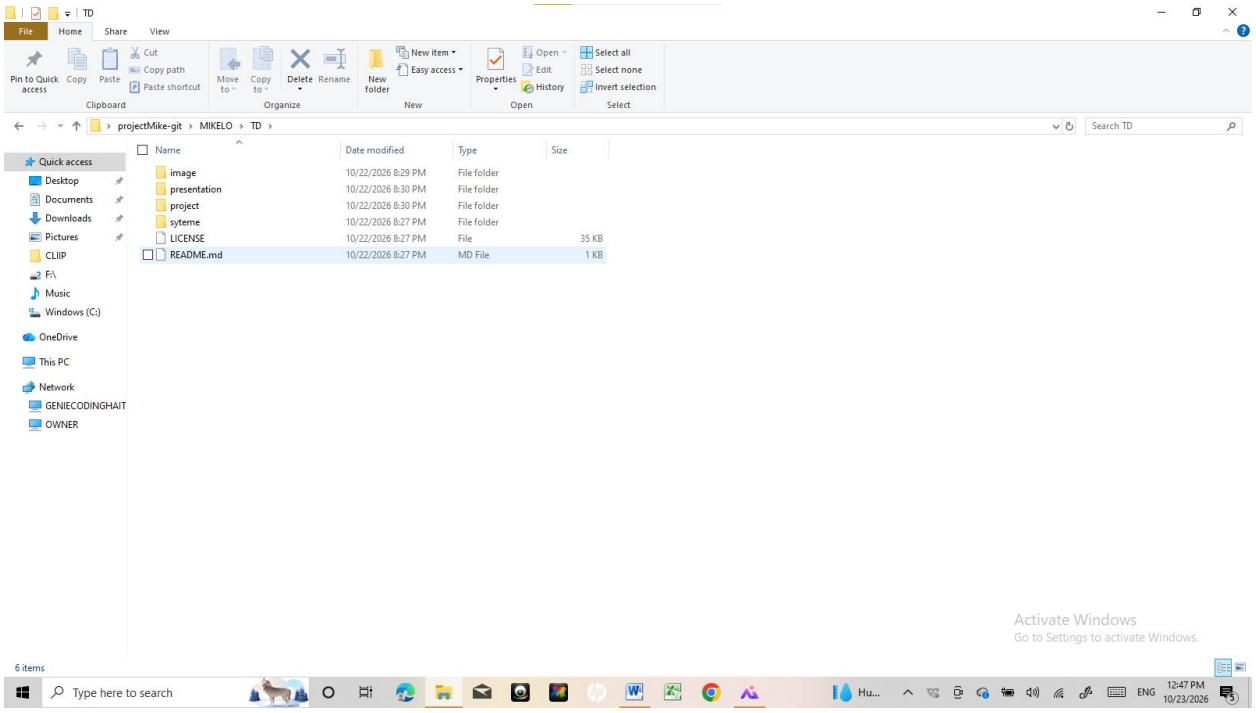
mkdir projet

mkdir presentation

cd ..


UCRT64:/c/Users/User/Desktop/projectMike-git/Mikelo
To github.com:Mikelo864/MIKELO.git
3c26fc6..9650016 main -> main
branch 'main' set up to track 'origin/main'.
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ cd TD
bash: cd: TD: No such file or directory
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ git clone git@github.com:Mikelo864/TD.git
Cloning into 'TD'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ cd TD
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/TD (main)
$ mkdir syteme
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/TD (main)
$ cd systeme
bash: cd: systeme: No such file or directory
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/TD (main)
$ mkdir image
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/TD (main)
$ mkdir project
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/TD (main)
$ mkdir presentation
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/TD (main)
$ cd ..
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ mkdir reseau1
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$ cd reseau1
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/reseau1 (main)
$ mkdir image
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/reseau1 (main)
$ mkdir project
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/reseau1 (main)
$ mkdir presentation
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo/reseau1 (main)
$ cd ..
User@Owner UCRT64 ~/Desktop/projectMike-git/Mikelo (main)
$
Activate Windows
Go to Settings to activate Windows.
Type here to search
Mait...
ENG
8:33 PM
10/22/2026


MIKELO
File
Home
Share
View
Pin to Quick access
Copy
Paste
Cut
Copy path
Paste shortcut
Clipboard
Move to
Copy to
Delete
Rename
Organize
New item
Easy access
New folder
New
Properties
Open
Edit
History
Open
Select all
Select none
Invert selection
Select
projectMike-git › MIKELO ›
Search MIKELO
Quick access
Desktop
Documents
Downloads
Pictures
CLIIP
F:\
Music
Windows (C:)
OneDrive
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Windows (C:)
RECOVERY (D:)
Network
Name
Date modified
Type
Size
reseau1
10/22/2026 8:32 PM
File folder
TD
10/22/2026 8:30 PM
File folder
Module
10/22/2026 8:11 PM
Text Document
1 KB
README.md
10/22/2026 8:03 PM
MD File
1 KB
Activate Windows
Go to Settings to activate Windows.
4 items
Type here to search
26°C
ENG
8:37 PM
10/22/2026



Conclusion

Notion acquises durant ce laboratoire :

- ✓ Familiarisation avec les termes Git (dépôt) et Github (dépôt distant)
- ✓ Capacité à héberger un projet sur Github et à cloner un projet sur notre ordinateur avec git
- ✓ Utilisation de certaines commandes Git et leurs utilisations (et d'autre termes techniques) :
    1. git --version : Permet de vérifier la version git installé sur son ordinateur.
    2. git config --global user.name et git config --global user.email : Permet de configurer le compte github avec des données utilisateurs.
    3. ls : Permet l'affichage de tous les répertoires
    4. git config --list : Permet d'afficher la liste des configurations de Git.
    5. cd : Permet l'ouverture d'un dossier.

mkdir : Permet de créer un dossier.

6. git init : Permet l'initiation d'un dépôt
7. README.md : Fichier texte écrit en Mark down.
8. git log : Permet d'afficher l'historique des commit dans un dépôt git.
9. git status : Sert à afficher l'état actuel du dépôt Git .
10. git add : Sert à ajouter des fichiers.
11. git commit –m : Valide les changements ajoutés.
12. git branch –M main : Permet de renommer la branche actuelle en main, même si une branche main existe déjà.
13. git push –u origin main : Permet d'envoyer la branche main vers Github.